Compiler Design

CSE - 420

Assignment 03

Spring 2026

Note that:
1. You will have to submit the hard copy
2. Make an cover page which contains you name,student ID and section

Deadline: 23 April, 2026(Class Time)

Draw the annotated parse tree for the input string using the attribute grammar given below. Then you have to compute the values of the type, width, and offset attributes for each variable following the SDT rules given in the attribute grammar and draw the attribute dependency arrows in your parse tree.

Input String: int [10] [20]a;

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record{
        float b;
        int [3] ab;
        }recor;
float abc;
```

P → { offset = 0; }D
D → T id ; {top.push(id.lexeme, T.type, offset); offset = offset + T.width; } D_1
D → ε
T → B { t = B.type; w = B.width; } C {T.type = C.type; T.width = C.width; }
T → record '{' { Env.push(top); top = new Env(); Stack.push(offset); offset = 0; } D '}' {T.type = record(top);
T.width = offset;
top = Env.pop(); offset= Stack.pop(); }
B → int { B.type = integer; B.width = 4; }
B → float { B.type = float; B.width = 8; }
C → ε { C.type = t; C.width= w; }
C → [num] C_1 {C.type = array(num.value, C_1.type); C.width= num.value * C_1.width; }